
Part 1: Universal Hadoop Installation & Setup Guide

Run these commands step-by-step to build a fresh, single-node Hadoop cluster from scratch.

1. Prerequisites & User Creation

Run as your default Ubuntu user.

```
sudo apt-get update
sudo apt-get install default-jdk ssh -y
sudo addgroup hadoop
sudo adduser --ingroup hadoop hduser
sudo adduser hduser sudo
su hduser
cd ~
```

2. Passwordless SSH Setup

Run as hduser. Press Enter if prompted to confirm connections.

```
ssh-keygen -t ed25519 -P "" -f ~/.ssh/id_ed25519
cat ~/.ssh/id_ed25519.pub >> ~/.ssh/authorized_keys
chmod 0600 ~/.ssh/authorized_keys
ssh localhost
exit
```

3. Download & Extract Hadoop

Run as hduser.

```
wget https://archive.apache.org/dist/hadoop/common/hadoop-3.3.4/hadoop-3.3.4.tar.gz
sudo tar xvfz hadoop-3.3.4.tar.gz
sudo mkdir -p /usr/local/hadoop
cd hadoop-3.3.4
sudo mv * /usr/local/hadoop
cd ~
```

```
sudo chown -R hduser:hadoop /usr/local/hadoop
```

4. System Environment Variables

Run as *hduser*. Open the file:

```
nano ~/.bashrc
```

Scroll to the bottom and paste:

```
#HADOOP VARIABLES START
export JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64
export HADOOP_INSTALL=/usr/local/hadoop
export PATH=$PATH:$HADOOP_INSTALL/bin
export PATH=$PATH:$HADOOP_INSTALL/sbin
export HADOOP_MAPRED_HOME=$HADOOP_INSTALL
export HADOOP_COMMON_HOME=$HADOOP_INSTALL
export HADOOP_HDFS_HOME=$HADOOP_INSTALL
export YARN_HOME=$HADOOP_INSTALL
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_INSTALL/lib/native
export HADOOP_OPTS="-Djava.library.path=$HADOOP_INSTALL/lib"
#HADOOP VARIABLES END
```

Save (*Ctrl+O*, *Enter*, *Ctrl+X*) and apply:

```
source ~/.bashrc
```

5. Configure Hadoop Files

Run as *hduser*. Edit the following 5 files located in */usr/local/hadoop/etc/hadoop/*.

A. `hadoop-env.sh`

```
sudo nano /usr/local/hadoop/etc/hadoop/hadoop-env.sh
```

Add to the file: `export JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64`

B. `core-site.xml`

```
sudo mkdir -p /app/hadoop/tmp
sudo chown hduser:hadoop /app/hadoop/tmp
sudo nano /usr/local/hadoop/etc/hadoop/core-site.xml
```

Paste between `<configuration>` tags:

```
<property><name>hadoop.tmp.dir</name><value>/app/hadoop/tmp</value></property>
<property><name>fs.default.name</name><value>hdfs://localhost:54310</value></property>
```

C. mapred-site.xml

```
sudo nano /usr/local/hadoop/etc/hadoop/mapred-site.xml
```

Paste between <configuration> tags:

```
<property><name>mapred.job.tracker</name><value>localhost:54311</value></property>
```

D. hdfs-site.xml

```
sudo mkdir -p /usr/local/hadoop_store/hdfs/namenode
sudo mkdir -p /usr/local/hadoop_store/hdfs/datanode
sudo chown -R hduser:hadoop /usr/local/hadoop_store
sudo nano /usr/local/hadoop/etc/hadoop/hdfs-site.xml
```

Paste between <configuration> tags:

```
<property><name>dfs.replication</name><value>1</value></property>
<property><name>dfs.namenode.name.dir</name><value>file:/usr/local/hadoop_store/hdfs/name
node</value></property>
<property><name>dfs.datanode.data.dir</name><value>file:/usr/local/hadoop_store/hdfs/datanod
e</value></property>
```

E. yarn-site.xml

```
sudo nano /usr/local/hadoop/etc/hadoop/yarn-site.xml
```

Paste between <configuration> tags:

```
<property><name>yarn.nodemanager.aux-services</name><value>mapreduce_shuffle</value></pr
operty>
<property><name>yarn.nodemanager.env-whitelist</name><value>JAVA_HOME,HADOOP_COMM
ON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF_DIR,CLASSPATH_PREPEND_DISTCACHE,H
ADOOP_YARN_HOME,HADOOP_MAPRED_HOME</value></property>
```

6. Cluster Ignition

Run as `hduser`. **CRITICAL: ONLY RUN THE FORMAT COMMAND ONCE!**

```
hdfs namenode -format
start-dfs.sh
start-yarn.sh
jps
```

(Verify you see: `NameNode`, `DataNode`, `SecondaryNameNode`, `ResourceManager`, `NodeManager`).

Part 2: Hadoop Troubleshooting Guide

If a command fails or `jps` is missing services, find your error below and run the fix.

Error 1: "0 datanodes running" OR `DataNode` missing from `jps`

Cause: You ran `hdfs namenode -format` more than once. The `NameNode` got a new ID, but the `DataNode` kept the old one, causing a mismatch crash.

The Fix (The Clean Reset):

```
# 1. Stop all services
stop-all.sh
```

```
# 2. Delete all corrupted storage data
sudo rm -rf /usr/local/hadoop_store/hdfs/namenode/*
sudo rm -rf /usr/local/hadoop_store/hdfs/datanode/*
sudo rm -rf /app/hadoop/tmp/*
```

```
# 3. Format ONE MORE TIME
hdfs namenode -format
```

```
# 4. Restart and check
start-dfs.sh
start-yarn.sh
jps
```

Error 2: "Permission denied (publickey, password)"

Cause: SSH requires a password, but Hadoop needs to connect automatically in the background.

The Fix:

```
# 1. Delete old keys
rm ~/.ssh/id_ed25519*
rm ~/.ssh/authorized_keys

# 2. Create a new key with NO passphrase
ssh-keygen -t ed25519 -P "" -f ~/.ssh/id_ed25519
cat ~/.ssh/id_ed25519.pub >> ~/.ssh/authorized_keys
chmod 0600 ~/.ssh/authorized_keys

# 3. Test it (should log in instantly without a password prompt)
ssh localhost
exit
```

Error 3: "Command not found" (e.g., start-dfs.sh: command not found)

Cause: Ubuntu doesn't know where the Hadoop /sbin folder is, usually because .bashrc wasn't sourced or has a typo.

The Fix:

```
# 1. Reload the environment variables
source ~/.bashrc

# 2. If it still fails, check for typos in your variables
nano ~/.bashrc
```

Error 4: "NameNode is in safe mode"

Cause: Hadoop shut down unexpectedly and locked the file system to protect data. You cannot upload files or create directories.

The Fix:

```
# Force Hadoop to leave safe mode
hdfs dfsadmin -safemode leave
```

Error 5: "File /user/hduser/output already exists"

Cause: You are running a MapReduce job, but you pointed it to an -output folder that was created by a previous test. Hadoop will not overwrite old folders; it crashes instead.

The Fix:

```
# Delete the old output folder from HDFS before running your job  
hdfs dfs -rm -r -f /user/hduser/target_output
```

Error 6: "Unable to load native-hadoop library"

Cause: A harmless warning stating that Hadoop is using standard Java libraries instead of C++ libraries for your specific processor architecture.

The Fix:

Ignore it. This warning will not stop your cluster or your practicals from running perfectly.